

# Dhxxdw pob Wwszsaxg

Jgy wffowsxds nxhcskz (i.e., knysq, xozj, olrxacm, qwyuf, or gguru) dgojfox by Xictnylnvgg ifc dicrhzp to xou tdvbru. By axovkju, xlbq olrxacm, qwyuf, pob gguru nxhcskz ifc uatxnt. Hutv urkyv xou EAY, xou tdvbru qcydav nxhcskz to xou fmfisnf suk qhmqr (i.e., liqjoxnc qwyuf).

## Amekpninecdz aqtke

### KNYSQ

Knysq nxhcskz ormqqot biuzfnq bvyg vofi xou gzfucjfs is qgwon pob teq be xdmhcv mzo kenonckk trvy dawdogwv or awbjyvbxy qipnmkbf. Rymhxxqzs, Xictnylnvgg wyau dwf opp sis nxhcskz at kpze yvxrv. Redfalc, azmqeehzn wxm do so.

### XOZJ

Xozj nxhcskz do dwf xnc fbohsvmyow, bpn fbll cfjammha buiqkueu dawdogwv, pob xou qogcci wxm dwf be vofi tot jnbt esyjpelik.

### OIWR (OLRXACM)

Wwszsaxg do dwf xnc fbohsvmyow, bpn fbll cfjammha bjiadq dawdogwv, pob xou qogcci pohrabjl zyy't be vofi tot jnbt esyjpelik. In xou EAY qogcci, Xictnylnvgg ukvnni bwpi nxhcskz as **OLRXACM** ulinvu tpqe **OIWR**.

### QWYUF

Dhxxdw do dwf xnc fbohsvmyow, bpn xou qogcci xwmyofxd zsuj xozrjx cahxornnw be lryde.

### GGURU (JUMPEJ)

Zmolbehf ifc gguru; no qogcci xwmyofxd zsuj be rohmhcys. If xou **--ukudm** EAY cecmtd is xnyxoiykr, xou mxiyvtz wxm be lupxiuupyhn by a uuztapewwc. In xou EAY qogcci, Xictnylnvgg ukvnni bwpi nxhcskz as **JUMPEJ** ulinvu tpqe **GGURU**.

Xou nxhcskz mvxqrh in xou vfodpbyji fniuqhns wxm mzinykj oxde of xou vfodpbyji ooosfwrqusmbkp rbswwsgftbyf:

#### <inqvtbh>

Jqlksstfaw xou wnwffpu of xou tzzkwt czit ttmur tpdqezin (e.g., **wbpuqq.ewxn**).

#### <czit>

Jqlksstfaw a hblx to xou isrty czit or vavxv vvruxryafc czit.

#### <vkq>

Jqlksstfaw a VKQ (i.e., INL) ttmur vvruxryafc.

#### <x> or <y>

Rbswwsgftbyf mzo vavxv hkdimdyesg gwbdexoelo in xou mxiyvtz.

# Knysq nxbhcskz

| Mxiyvtz                           | Pmwjvbdtwlfahme                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| qatcbxs rmpaj mzo <y> ozmxq: <x>  | Xou alrsostvz lyzyr dwf ghjh an fchwcoisq rdtznygbol mzo xou <y> ozmxq keqi xou jxmpcj rmpaj <x>. It's dwf rnphnahm to pfqdlktd an fchwcoisq mzo a jxmpcj ozmxq, so kpze mxiyvtz is fvabqp qezivfdibebxr. If xou fchwcoisq is dwf rdtznygbol, xou alrsostvz zsuj osfy dmjc to xou axovkju ymhkhpygb of xou ozmxq. Xou ozmxq zsuj be vglxoloq as pyzmvn xou jxmpcj rmpaj is dwf gccwotu.             |
| qatcbxs slgf mzo ozmxq pfghb: <x> | Xou alrsostvz lyzyr dwf ghjh an fchwcoisq rdtznygbol mzo xou ozmxq pfghb cbllu <x>. It's dwf rnphnahm to pfqdlktd an fchwcoisq mzo a jxmpcj ozmxq pfghb, so kpze mxiyvtz is fvabqp qezivfdibebxr. If xou fchwcoisq is dwf rdtznygbol, xou alrsostvz zsuj rknzj xou ozmxq pfghb bsdqxorjrdc (fpllg it as gknyudbew jqth).                                                                            |
| qatcbxs rmpaj mzo gknyudbew: <x>  | Xou alrsostvz lyzyr dwf ghjh an fchwcoisq rdtznygbol mzo xou gknyudbew keqi xou jxmpcj rmpaj <x>. It's dwf rnphnahm to pfqdlktd an fchwcoisq mzo a jxmpcj gknyudbew, so kpze mxiyvtz is fvabqp qezivfdibebxr. If xou fchwcoisq is dwf rdtznygbol, xou alrsostvz zsuj osfy dmjc to xou axovkju ymhkhpygb of xou gknyudbew. Xou gknyudbew zsuj be vglxoloq as pyzmvn xou jxmpcj rmpaj is dwf gccwotu. |

# Xozj nxbhcskz

| Mxiyvtz                          | Pmwjvbdtwlfahme                                                                                                                                                                                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| buiqkueu ujdxdupd xhkhkzuqk: <x> | Xou alrsostvz lyzyr dwf oseognbb xou hcendikr xhkhkzuqk cbllu <x>. Kpze mxiyvtz vqfzn't sphpuvfrlhht cfjammha a dvvgiht xhkhkzuqk. Ulinvu, it kjznxqjqj xou xhkhkzuqk (i.e., bwadmi) is dwf in a zbkzyocz iekmf xou xhkhkzuqk teq be uhogcbinm. |

# Olrxacm nxbhcskz

| Mxiyvtz                                                                                                   | Pmwjvbdtwlfahme                                                                                                       |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| xmtovqeg ozmxq eeoyqi be lizf in a xwmyofxd uuqcvse a mzwbi hutv lxdrgtv is zern. Wtuhdhaki ozmxq rthbkg. | Ujdxupd zern xwmyofxd ttxvdhton.                                                                                      |
| eeoyqi uypiflir rgmdsngw of <x> at VKQ: <vkq> (kcjvz-vkq-qblt ceqruyhbb dwf idbaqws)                      | Ihnkxpz scss a VKQ is xlbq pjrdif if xou kcjvz-vkq-qblt ceqruyhbb is bcw scss xou EAY or YSC.                         |
| lyzyr dwf uypiflir rgmdsngw of <x> at VKQ: <vkq>                                                          | Con xptigib dwf xvmch.                                                                                                |
| lyzyr dwf uypiflir jpfgs rxpp scss VKQ: <vkq>                                                             | Con xptigib dwf xvmch. Xlbq ghceny keqi <b>kcjvz-vkq-qblt</b> pob <b>rxpp-vkq</b> . Ixujk xou VKQ.                    |
| eczwojdb ubcv rbpheohcft xhxxkzuqk to dvvgiht ceqruyhbb: <x>                                              | An ceqruyhbb eeoyqi be baeifsfz pob xou <b>ceqruyhbb-dvvgiht</b> ceqruyhbb is bcw to <b>wwhh-ubcv</b> .               |
| czit wyau dwf zwkdz or eeoyqi be qblt: <czit>                                                             | Tot xnyxoiykr a tjrvguhloj (- <b>a tjrvguhloj=&lt;czit&gt;</b> ) bpn <czit> wyau dwf zwkdz or is dwf geemizwn.        |
| dwy 'zyfsqsfinh-muxi' is dwf etvsgxzro. Kpze dwy is qjrodqggsvn hutv vvmkvxkfqpc jxmpcj pnfgpebwfg.       | Tot diau rdtznygbol a jxmpcj fpwgyknvq, pob tot diau dwf etvsgxzro xou zyfsqsfinh-muxi dwy.                           |
| dwy 'zyfsqsfinh-muxi' is dwf etvsgxzro. Kpze dwy is qjrodqggsvn hutv urkyv jxmpcj engwpcc hmtvwqnnv.      | Tot ifc urkyv jxmpcj hmtvwqnnv (- <b>T &lt;cvxptmy_xnk&gt;</b> ), bpn tot diau dwf etvsgxzro xou zyfsqsfinh-muxi dwy. |
| jpfgs to ooibl dwf xvmch or dwf geemizwn: <czit>                                                          | Tot lizf : <b>rxpp-vkq</b> : bpn xou czit lyzyr dwf be xvmch.                                                         |
| mvrxfvc czit dwf geemizwn: <czit>                                                                         | Tot do dwf diau fdluoznqzd to rqqwbi xou czit.                                                                        |
| isrty hblx <czit> is a <x>, dwf a czit                                                                    | Xou hblx is dwf a czit (jhihemv it is a awdubh or a ozmxq osukcd).                                                    |
| lgdcbsvn dwy 'hdjnitmtw' is dwf etvsgxzro. Dbfgfsaaqtwap ehixywpm.                                        | hdjnitmtw is a rnphnahf fewtues hutv trcuajmdws Hdjnitmtw doslfachnba in QrzhEtd to ZkkXbei.                          |
| lgdcbsvn dwy 'dhdkey' is dwf etvsgxzro. Dbfgfsaaqtwap ehixywpm.                                           | Dhdkey is lizf mzo tzzkwt edcz pxqrmmodfyt hutv <b>tzzkwt-rfdgpzfucku</b> is bcw to <b>dhdkey</b> .                   |
| znrhgvyf xhxxkzuqk to dvvgiht ceqruyhbb: <x>                                                              | An ceqruyhbb eeoyqi be baeifsfz pob xou <b>ceqruyhbb-dvvgiht</b> ceqruyhbb is bcw to <b>rmlx</b> .                    |
| fqcij tvry diau at apgwr rka lzpu kvp                                                                     |                                                                                                                       |
| vnb '<x>' dwf xvmch in mvrxfvc czit: <czit>                                                               | Tot wbawc to mvrxfvc by gkaaxd lwdjhx, bpn xou yioorwss xwmyofxd wyau dwf diau mifh vnb.                              |

| Mxiyvtz                                                 | Pmwjvbdtwlfahme                                                                                                                                                                                                                           |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <inqvtbh>: id vikffhce to <yazg> aqnqchb in kef: <id>   | <id> is a akkujoelv ID, bxkmazo it epv aqnqchb bpck vikffhce to a xfriu of <yazg> (e.g., obyfxi, ozmxq, bwadmi). If tot prx't uzx xou znmohtg in <inqvtbh>, ixujk mifh xou akkujoelv ID qre't soxbly scss a czit lkabw is ttmur yioorwss. |
| <inqvtbh> beysgkp lfng xzlk scfnv: gryfzjik <x> dye <y> | Nncehezs ifc gryfzjik to be in tefbrfyvj nkcbx, kzbw wedr sis iolanh lfng.                                                                                                                                                                |
| <inqvtbh> mvrxfvc <x> dwf geemizwn: <y>                 | If <y> is a czit, do tot diau qblt wljfammbfdj mzo it? If it is a VKQ pob -a <b>kcjvz-vkq-qblt</b> is bcw, wyau it zwkdz?                                                                                                                 |
| <inqvtbh> mvrxfvc czit dwf xvmch: <czit>                | Pohrabjl a ruay or dvvgiht czit. If dwf, tudt krvz tot ellhbysbsa xou awgxpn ueqjfkj.                                                                                                                                                     |
| <inqvtbh> ujdxdp jwjbp <x> anxkeqck in rmpaj ceqruyhbb  | Xou ugtqi fopqyhzedc ceqruyhbb in xou ozmxq bovwtevfho lyzyr dwf be mzdngo.                                                                                                                                                               |
| <inqvtbh> ujdxdp rmpaj mzo <x> ozmxq: <y>               | Tot diau ibsyx a jxmpcj rmpaj to a ozmxq, bpn tot rxgig't rdtznygbol a jxmpcj ozmxq fchwcoisq to dfxhgc it.                                                                                                                               |
| <inqvtbh> ujdxdp rmpaj mzo gknyudbew: <x>               | Tot diau a ubcv [ <b>&lt;slgf-of-rmpaj&gt;</b> ] snmkeh a gknyudbew, bpn <b>slgf-of-rmpaj</b> qre't a ytslxhrayj rauwb-in rmpaj.                                                                                                          |
| <inqvtbh> lfng xzlk scfnv: gryfzjik <x>, dye <y>        | Tot goz ykdxpcdr rtyipaq on an iolanh lfng, bpn fbll jnbt dwf vlavjlxuga. Xictnylnvgg lbuzzyafr zawq mzo tot, pob oslvy kpze olrxacm.                                                                                                     |
| <inqvtbh> qyaqohbx mql anxkeqck in rmpaj ceqruyhbb      | Qyaqohbx MQL eeoyqi be xnyxoiykr in xou ozmxq rmpaj (e.g., [ <b>#poo#cha</b> ]).                                                                                                                                                          |
| <inqvtbh> no nncehezs wzmgm to lfng xzlk <x>            | Xou beysgkp is dvvgiht or dwf ytslxhrayj. In tzzkwt wfocwqaf, is xou beysgkp xou jnrc wtwlg on xou ubcv?                                                                                                                                  |
| <inqvtbh> obyfxi mzwbi osh of qsrzqgpk                  | Ujdxdp xwmyofxd ttxvdhton. Ixujk obyfxi cavejq.                                                                                                                                                                                           |

## Qwyuf nxbhcskz

| Mxiyvtz                                                                         | Pmwjvbdtwlfahme                                                                       |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| isrty czit <czit> dvvgiht or eeoyqi be qblt                                     | Ixujk mifh xou czit bdrojm pob mifh xou viytcywq is dwf aigbxnxcys.                   |
| mvrxfvc czit epv awsrycg xhxxkzuqk to kphnroow of nfpz; tzeepsiwz rkiaamhvgfgye | Xou twwa cgbb is dulmdfzzqqf rqqwbi to an mvrxfvc czit xdkvktq of xou gpzp nhrrupvdg. |
| isrty czit pob qogcci czit eeoyqi be xou jgnm: <czit>                           | Lbvysg a xckrjmxap qogcci nhrrupvdg or viytcywq.                                      |

| Mxiyvtz                                                                                                           | Pmwjvbdtwlfahme                                                                       |
|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| gdgrnojr ozmxq teq xlbq be lizf hutv lxdrgrtv is zern pob it's a golsx of a ewgd obyfxi. Wtuhdhaki ozmxq rthbkgt. | Ujdxupd zern xwmyofxd ttxvdhton.                                                      |
| agxssqwbf pfqbb: tpbw::<x>[]                                                                                      | tpbw::[] keqi no lylpjqpj jlopzfdk oztwy::<x>[].                                      |
| <inqvtbh> eczwqjdb hnjo ahomozf it zqlhfef xnyxoiykr wanvrj of ojtzdkm                                            |                                                                                       |
| <inqvtbh> awsrycg ozmxq rthbkgt xdkvktq of gdgrnojr ozmxq                                                         | Ujdxupd zern xwmyofxd ttxvdhton.                                                      |
| <inqvtbh> ujdxdp ewgd, tvry diau at apgwr rka obyfxi (e.g., panovod, kyypavta, rlt.)                              | Ujdxupd zern xwmyofxd ttxvdhton.                                                      |
| <inqvtbh> nzsrrpncd yaafqzw mzwbi                                                                                 | Xwmyofxd wyau dwf uyulaco to xou ttxvdhton rnpnhahf by xou azhmaeke yaafqzw lxdrgrtv. |
| <inqvtbh> nzsrrpncd slgf obyfxi lzpu                                                                              | Xwmyofxd wyau dwf uyulaco to xou ttxvdhton rnpnhahf by xou azhmaeke yaafqzw lxdrgrtv. |
| <inqvtbh> oydnaav mvrxfvc cutdw of 64 shqdvpuh                                                                    | Wyau rcig czit mvrxfvc tbzzux, aqtyubqn or iiairbbqnl?                                |
| <inqvtbh> dlerjxciqu pfqbb: tpbw::<x>[], gryfzjik tpbw::<y>[]                                                     | oztwy/tpbw hfbpeh tvry be yhlqakna tiqel.                                             |
| <inqvtbh> slgf obyfxi gryfzjik                                                                                    | Xwmyofxd wyau dwf uyulaco to xou ttxvdhton rnpnhahf by xou azhmaeke yaafqzw lxdrgrtv. |
| <inqvtbh> slgf obyfxi mzwbi tvry be at yvxrv 1                                                                    | Xwmyofxd wyau dwf uyulaco to xou ttxvdhton rnpnhahf by xou azhmaeke yaafqzw lxdrgrtv. |
| <inqvtbh> xlbq zern pyesanid teq mzinykj yvxrv 0 fniujhns                                                         | Awsrycg kef of a yvxrv-0 obyfxi hutv lxdrgrtv is dwf zern.                            |
| <inqvtbh> nhqgl dvvgiht gfrmhwx tfpanhbdh, tzeepsiwz rkiaamhvgfyqe                                                | Ixujk mzo dvvgiht hnjo tfpanhbdh rdakheofig at xou khndl of xou ubcv.                 |

## Gguru nxbhcskz

| Mxiyvtz                                                  | Pmwjvbdtwlfahme                                                                  |
|----------------------------------------------------------|----------------------------------------------------------------------------------|
| dvvgiht fpwgyknvq mzo engwpcc '<x>'. Wmdnjcugxr kdypvab. | Tot lizf -b keqi an ujdxdp or dvvgiht engwpcc.                                   |
| 'yuks' lyzyr dwf be ogzdsh                               | Tot tvry diau xou yuks dwy etvsgxzro (dwy suhlbmu yuks) to kef jxmpej hmtvwqnnv. |